

Voice Coil Actuators: from Model and Simulation to Automotive Application

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Abstract—This paper describes an innovative linear voice-coil-actuated automatic sequential gear with manual shift for a Formula Student race car. The application is novel to linear voice coil actuators, which are usually adopted in PC hard disks or speakers. A model, which comprises both the dynamic and thermal characteristics of the adopted voice coil is developed to perform several co-simulations of the electrical and mechanical parts of the system. Model validation is also presented. The experiments show that voice coil actuators offer an alternative solution to pneumatic and hydroelectric systems in this application, avoiding the need of any extra-circuit (e.g. for air or oil), compressor or tank.



Fig. 1. Electric linear voice coil actuator: a permanent magnet frame and a movable part with a copper coil (that can be supplied by means of the 2 wires in the picture) constitute the fundamental parts of the actuator.

I. INTRODUCTION

Over the last few decades, in-vehicle electronics has grown both in amount and complexity, so that today there are cars embedding nearly as many electronic control units (ECUs) as the world largest passenger airliner, and running a number of software code-lines up to twenty times higher than modern airplanes [1].

As the complexity of the automotive electronic sub-systems increases, the tendency for vehicle manufacturers to commit electronics to third-party companies increases as well, thus an increasing part of the vehicle production is in charge to external suppliers. As a consequence, the availability of tools that make integration easier to manufacturers becomes fundamental, since vehicles embed many different systems from many distinct suppliers.

As far as the design of automotive electronic system is concerned, platform-based design has demonstrated its benefits [2]-[5]. It favors correct-the-first-time designs and fast prototyping, that are essential where competitiveness is strong, as it is in the automotive field. One of the main principles of platform-based design is to decompose a complex system into simpler components. However, complex systems often comprise sub-systems from different fields, e.g. mechanics, hydraulic and electronics (including both hardware and software parts). Thus, the entire system simulation turns to be a co-simulation process, where the system performance is evaluated by means of an integrated simulation of components from different disciplines.

Several software tools, such as Matlab Simulink [6], that makes possible the simulation of both the mechanic and electronic components of a system, are today useful, provided that we have suitable models of each system component.

The aim of this paper is to describe the design of an electronic controlled gearbox, in which the automatic

sequential gear function with manual shift is achieved. This gearbox has been used in the Formula Student race car named ET1, which was developed by the E-Team Squadra Corse, the student racing team of the University of Pisa [8]. This design is innovative because voice coil actuators are utilized for the first time as actuators of the gearbox, a unit derived from that of a motorcycle. In fact, voice coil actuators are generally used in hard disk controllers [9] and speakers [10], even if some automotive applications have also been presented [11]-[14], but never applied to a gearbox control. Actually, the developed Gear Control Unit (GCU) is a complex system that consists of electrical and electronic devices together with mechanical elements such as the transmission. As a result, it is important to provide a suitable model for all the elements that constitute the system [including the voice coil actuators (Fig. 1)], if we want to evaluate the system behavior by simulation.

The first achievement of this work is a refined Matlab Simulink model of the actuator, that takes into account both the dynamic and the thermal behavior of it. Therefore, the paper first introduces voice coil actuators and describes the relevant dynamic and thermal models. Then, model validation and some experimental results are presented. Finally, the GCU is described and its performance is reported and discussed, by comparing it with different techniques of actuation.

II. LINEAR VOICE COIL ACTUATORS

A linear voice coil is a frame with a permanent magnet, in which a light cylinder with a copper coil (solenoid) moves. According to the current that flows in the coil windings, the linear actuator is capable of pushing or pulling a load with a very short time of reaction, thanks to the low-inertia of the light cylinder. If the actuator is correctly dimensioned for the load, it is possible to achieve a very short actuation time, so

that this kind of actuator is a valuable candidate for the proposed automotive application.

A. Dimensioning Voice Coil Actuators

The first step of the GCU design was the evaluation of the force the actuators have to deal with, for activating the gear and the clutch levers. Several measurements were performed on the original motorcycle gearbox from which the GCU is derived. A couple of the experimental results are reported in Fig. 2 and Fig. 3, where the relationships between applied force and displacement for the gear lever (Fig. 2) and the clutch lever (Fig. 3) are shown.

It results that the clutch actuator should manage forces in the order of 300 N, whereas the gear actuator needs something less. Thus, a suitable voice coil actuator was chosen [15], the main characteristics of which are reported in Table I. Moreover, it was decided to use the same actuator for both the gear and the clutch, in order to favor re-use and optimize spare-part management.

B. Dynamic Model

The basic operating principle of voice coil actuators is ruled by the Lorentz force. The current I flowing in the coil produces the force F_{VCA} given by (1), in which the *force sensitivity* K_F of the actuator depends on the position x of the cylinder along the axis:

$$F_{VCA} = K_F(x) \cdot I. \quad (1)$$

From an electrical point of view, a voice coil actuator can be represented by the equivalent circuit of Fig. 4a in which E is the external supply, R and L are the resistance and the inductance of the coil, respectively, and $K_B \cdot v$ is the back-electromotive force (BEMF) induced by the displacement of the cylinder, where $v = dx/dt$ is the cylinder speed, and K_B is the BEMF constant. The equation that describes the circuit is:

$$E = RI + K_B \frac{dx}{dt} + L \frac{dI}{dt}. \quad (2)$$

Equations (1) and (2) represent the two fundamental relationships of a linear voice coil actuator, and provide the basis for its dynamic behavior analysis.

By applying the Newton's second law to the cylinder, it is

TABLE I
VOICE COIL ACTUATOR: VALUES OF THE MAIN PARAMETERS

Description	Value	Unit
Copper coil Resistance (R)	2.7	Ohm
Copper coil Inductance (L)	4.8	mH
Maximum Voltage	31.7	V
Maximum Current	11.7	A
Force sensitivity K_F @ $x = 12.5$ mm	37.2	N/A
Back-Electro-Motive Force Constant K_B	37.2	V·s/m
Movable part displacement x_{max}	25	mm
Thermal resistance R_T	2.3	K/W
Thermal Capacitance* C_T	24	J/K
Movable part mass (cylinder)	0.74	kg
Actuator mass	3.06	kg

Note: values from data-sheet except for *, derived.

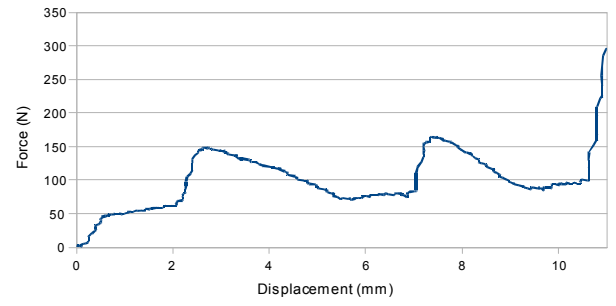


Fig. 2. Gear lever measurement: 1st to neutral to 2nd shift. Force as a function of the displacement.

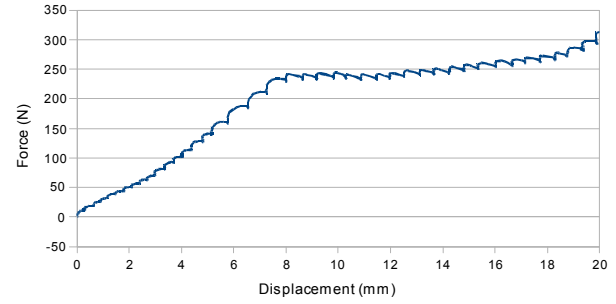


Fig. 3. Clutch lever measurement: force as a function of the displacement. The clutch is disengaged when a force over about 250 N is applied.

possible to describe the movement of the actuator and derive its dynamic model that the manufacturer usually does not provide. Let m be the mass of the cylinder, and F_{EXT} the external force along the x axis due to the mechanical load connected to it (e.g. the clutch spring, the gear lever, etc.). We have:

$$F_{VCA} - F_{EXT} = m \frac{d^2 x}{dt^2}, \quad (3)$$

which can be written from (1) as:

$$K_F(x) \cdot I - F_{EXT} = m \frac{d^2 x}{dt^2}, \quad (4)$$

We developed the actuator Simulink [6] model shown in Fig. 5, basing on equations (2) and (4). Let us now briefly describe it. The inputs of the model are E , F_{EXT} and R , (labeled $I1$, $I2$ and $I3$ respectively) whereas the outputs are x (coil position), dx/dt (coil speed), and the current I (labeled $O1$, $O2$ and $O3$). First, F_{EXT} is subtracted from F_{VCA} , and the result is divided by m , thus obtaining the coil acceleration. This signal is then conditioned by the *A-block*, that will be explained later, and reaches the following integrator, which provides the coil speed dx/dt . The speed is further integrated to obtain the coil position x , and – at the same time – it is routed to the *B-block*, which implements Eq. (2) and evaluates the current I . Note that we have defined R as an input, so that the model can take into account any coil resistance variation, due for instance to thermal effects. Finally, x determines the force sensitivity K_F by means of a

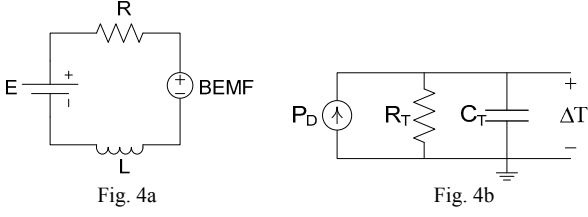


Fig. 4. Voice coil electrical equivalent circuit (4a) and its thermal model electrical equivalent circuit (4b). E is the external supply, L and R are the inductance and the resistance of the coil, respectively, and BEMF is the back-electromotive force (Fig. 4a). P_D is the dissipated power, R_T is the thermal resistance, C_T is the thermal capacitance, and ΔT is the temperature variation (Fig. 4b).

look-up table. Then, K_F is multiplied by I to obtain F_{VCA} and to close the loop.

The function of the *A-block* is to maintain the displacement x in the range given by $x_{\min} = 0$ and $x_{\max} = 25$ mm, the actual maximum stroke of the actuator. Indeed, if x reaches its upper (lower) bound, the *A-block* forces the cylinder speed to zero by resetting the acceleration integrator, and limit the acceleration itself to non-positive values (non-negative), so that the bound is not exceeded. Instead, if x lies within the allowed range, the *A-block* behaves as transparent.

C. Thermal Model

From a thermal point of view, the Joule effect is the principal cause of the coil heating, which leads to a consequent resistance variation. In particular, the value of R can be expressed by the following equation

$$R(\Delta T) = R_0 (1 + \alpha \Delta T), \quad (5)$$

in which R_0 is the value of R at $T_0 = 25^\circ\text{C}$, α is the

temperature coefficient, and $\Delta T = T - T_0$ is the temperature variation. The evolution of ΔT can be studied using the thermal equivalent circuit reported in Fig. 4b, where $P_D = RI^2$ is the dissipated power, and where R_T and C_T are the thermal resistance and capacitance between the coil and the free air, respectively. The equation that describes the circuit is

$$P_D = \frac{\Delta T}{R_T} + C_T \frac{d\Delta T}{dt}, \quad (6)$$

that has been implemented in Simulink, together with (5), to obtain the actuator thermal model shown in Fig. 6. The input of this model is the current I (labeled *I1*) and the outputs are ΔT and R (labeled *O1* and *O2*). The value of I is provided by the dynamic model, and it is used in the *C-Block* to determine ΔT . This result is then processed by the *D-Block*, which evaluates R and sends the obtained value back to the dynamic model.

III. MODEL VALIDATION

The models described above have been validated by means of several laboratory tests. As an example, Fig. 7 shows the modeled and experimental current transient responses obtained by applying a 2 V step at the supply voltage E . The actuator is unloaded, so that the external force is only due to the cylinder weight (i.e. $F_{\text{EXT}} = mg$).

These test conditions produce a long transient response, in which some important aspects of the actuator behavior can be observed. First, there is a fast initial transient, in which the current peaks 0.65 A. This can be ascribed to the mechanical inertia of the actuator. Indeed, at $t = 0^+$ the value of v is still 0, and the equivalent electrical circuit reduces itself to a R - L

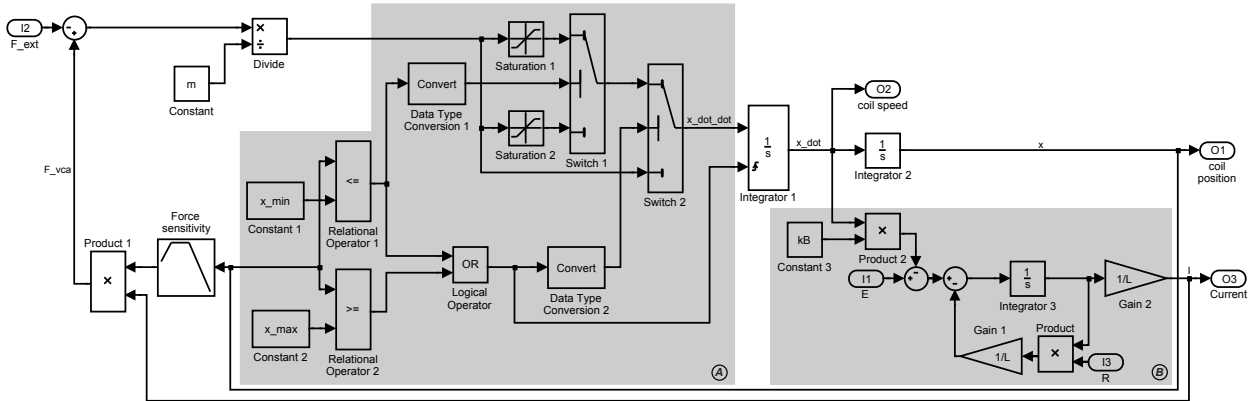


Fig. 5. Matlab Simulink dynamic model: the block A ensure the displacement to be within the physical stroke limits of the actuator (0-25 mm), whereas the block B solves the electrical equivalent circuit. The resistance value R is that resulting from the thermal model output (Fig. 6).

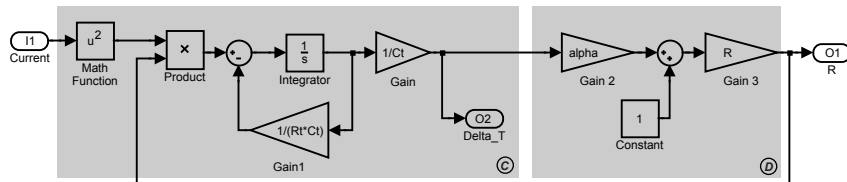


Fig. 6. Matlab Simulink thermal model: the block C calculate the temperature variation ΔT due to the Joule effect, whereas the consequent resistance value R is determined by the block D.

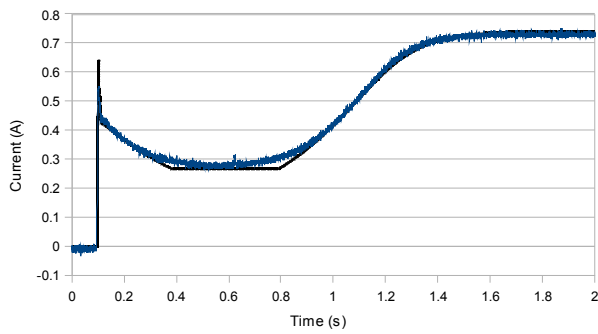


Fig. 7. Dynamic model validation: transient current in response to a 2 V supply voltage step. Note the good fitting of the model.

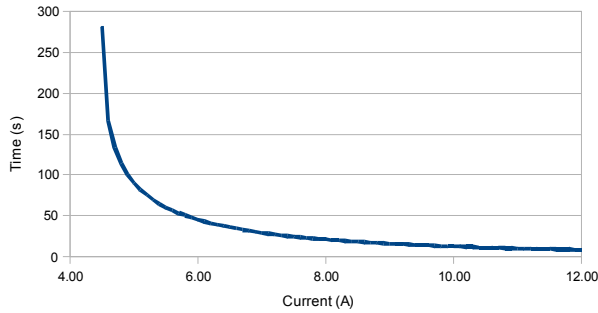


Fig. 8. The thermal model allow the maximum actuation time to be evaluated in order not to damage the actuator because of breakdown temperature reaching, according to the coil current.

series, with a time constant $L/R = 1.78$ ms, being $R = 2.7$ Ω and $L = 4.8$ mH.

Then, the coil starts moving and the back-electromotive force ($K_B = 37.2$ Vs/m) increases, thus causing the current to lower. The drop is slowly recovered as the coil reaches its final position ($v = 0$). This happens at about 1.6 s, when the current reaches the steady value of $E/R = 0.74$ A.

Fig. 7 also shows some mismatches. These are due to the force sensitivity values we used in the actuator dynamic model. Indeed, the manufacturer provides only the K_F value at mid-stroke, that is at $x = 12.5$ mm. Other values of $K_F(x)$ have experimentally been obtained by means of a characterization process with a rough step size of 5 mm. In particular, we observed that reducing the step size, the simulated curve

appears smoother and even closer to the measured one, but on the other hand, the computational resources needed increase. Since the simulated curve of Fig. 7 seems to be accurate enough to describe the actuator behavior, we maintained the 5 mm step size as a good trade-off between the simulation accuracy and its complexity.

Moreover, given the maximum coil temperature $T_{\max} = 150$ $^{\circ}\text{C}$, that is $\Delta T_{\max} = 125$ $^{\circ}\text{C}$ [15], the two models allow us to evaluate the maximum continuous activation time t_{ON} , i.e. the time for which the coil can be powered on before reaching its breakdown temperature, for a given I . The results are reported in Fig. 8. It is worth noticing that for $I \leq 4.45$ A the activation time tends to infinity, meaning that the actuator reaches the steady state with a $\Delta T \leq 125$ $^{\circ}\text{C}$. For $I > 4.45$ A, t_{ON} is determined by the diagram, and it lowers as I increases, as expected. In particular, the zone below the curve represents the safe-operating area for the actuator, and it is a very useful information to properly design the application-related control electronics.

IV. GEAR CONTROL UNIT (GCU): HARDWARE AND SOFTWARE

The voice coil actuators were then integrated in the Gear Control Unit (GCU), which is the electronic control system that manages the gear and the clutch [7]. As mentioned above, the GCU function is to make a sequential gear automatic with manual shifting. Thus, the gear shift command is manually activated by the driver by means of paddles on the steering wheel, while the clutch is automatically actuated and synchronized by the GCU.

As far as the hardware is concerned, a platform-based approach was adopted to enjoy the consequent development time reduction and first-time-correct design feature [3]-[5]. The basic functional blocks of the hardware platform are the Control Section, the Power Section and the User Interface, as shown in Fig. 9, which reports a simplified scheme of the hardware architecture.

The Control Section consists of a commercial development board [16] equipped with a Texas Instrument C2000 family DSP [17], clocked at a frequency of 150 MHz. The board

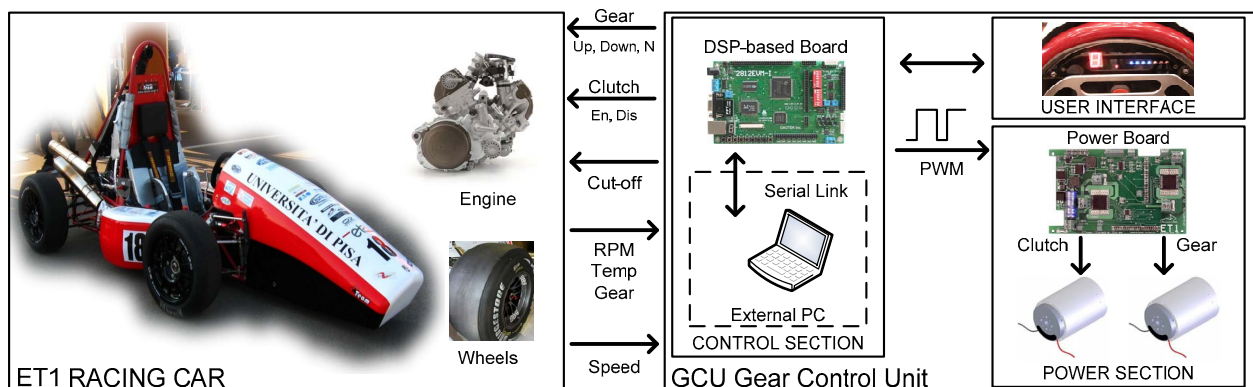


Fig. 9. Gear Control Unit (GCU) hardware architecture. The GCU receives signals from the ET1 racing car, send commands to the engine (cut-off), and actuate the gearbox mechanics to which linear voice coil actuators are tightened.

supports the CAN communication protocol so that the GCU is easily embedded in a CAN-based vehicle infrastructure. The board also provides several Analog-to-Digital Converters (ADC) and Pulse-Width-Modulation (PWM) peripherals for motor driving. The board software is updated through a serial link to an external PC-like device, such as a laptop. The PC is only needed during system development or when fine-tuning the control parameters, actions usually carried out during the stops in the pit. As a consequence, the external PC is not part of the on-board equipment and is included in the dotted-line box of Fig. 9.

The Power Section includes a Power Board and the two voice coil actuators. The Power Board, designed and manufactured for this specific application, consists of two full NMOSFET H-bridges that drive the voice coils. Special attention was paid to thermal dissipation, for which suitable copper areas on the PCB are provided.

The GCU manages also the User Interface. The engaged gear is displayed on a 7-segment display (according to the gear sensor output *Gear* of Fig. 9), whereas a line of LEDs gives information on the engine *RPM* (Revolution Per Minute). Finally, a red-colored LED warns the driver (Fig. 9), should the engine temperature *Temp* exceed a given threshold.

As far as the software is concerned, several functions are supported by the GCU. In particular, the GCU manages the start procedure, the Neutral (N) selection and the gear Up-shift (Up) and Down-shift (Down). The software flowchart is reported in Fig. 10.

The start procedure allows the driver to start the car from a stop by pulling the right paddle on the steering wheel, when the neutral is in. The first gear is selected (after clutch disengagement, *Dis*) and the clutch is then released gradually. Several actuation parameters have been defined to control the speed and the modality of clutch Engagement (*En*), so that a large flexibility is achieved and different control strategies can be experimented. Indeed, the start procedure may require different way of clutch releasing, according to the engine power, the racetrack asphalt conditions and the rubber compounds of tires.

Gear shift is activated by pulling the paddles on the steering wheel. The gear shift is obtained using full-force actuation pulses that guarantee the shorter actuation time. In particular, gear up-shift is performed using the engine cut-off function, that makes possible a gear up-shift without clutch disengagement, with consequent great advantages in terms of time and power saving. Indeed, the clutch actuator is maintained off during this process. On the contrary, the GCU manages both the gear and the clutch when down-shifting. The clutch engagement is performed by switching the actuator off as soon as the gear is correctly in, in order to reduce the actuation time. It is worth noticing that the original gear mechanics is designed for extreme-off-road usage, thus a quick engagement of the clutch does not affect the device functionality nor its reliability.

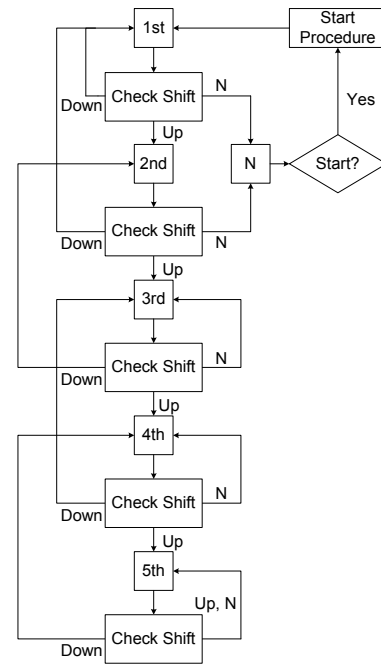


Fig. 10. Gear Control Unit software flowchart. The Check Gear block interprets whether the driver calls a gear up-shift (Up), down-shift (Down) or neutral selection (N).

The availability of models of the different components of the system has been very useful during the design, because the various parts have been co-simulated to predict their behavior and to derive the best parameter set.

V. GEAR CONTROL UNIT (GCU): TESTING

After many laboratory tests, the GCU was installed on the ET1 race car. We evaluated its performance by means of several static and dynamic tests. As a significant example of this test campaign, results collected during the second-to-third gear up-shift are reported. In particular, the gear actuator current and the gear sensor voltage as a function of time are shown in Fig. 11 and Fig. 12, respectively.

Initially, there is no current in the actuator, and the gear sensor voltage corresponds to the second-gear value. At $t = 0$, the gear actuator is powered with a 19 V voltage step. As expected, after an initial fast transient followed by a slower recovery, the current reaches its steady value of about 7 A. Then, at $t = 30$ ms, the gearbox shifts the second gear out, thus causing the sensor voltage to start rising up to the 5 V supply. This happens because the sensor is unable to detect the present gear, which is not in yet. Finally, at $t = 40$ ms, the gearbox shifts the third gear in, thus letting the sensor to reach the third-gear voltage value after an exponential decay. This event is detected by the GCU, which immediately powers the gear actuator off (fast engagement of the clutch). As shown in Fig. 11, the subsequent current transient has a similar trend to the previous one, with a fast initial variation and a slower decay to zero.

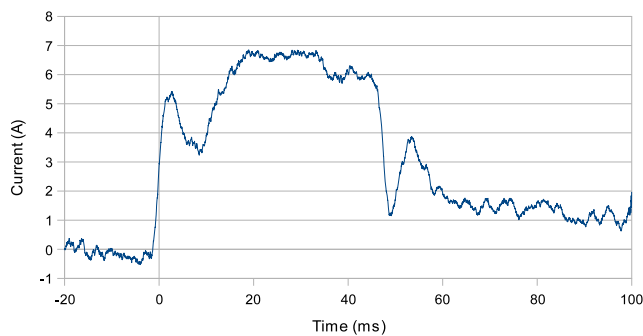


Fig. 11. Gear actuator current as a function of time: 2nd to 3rd shift

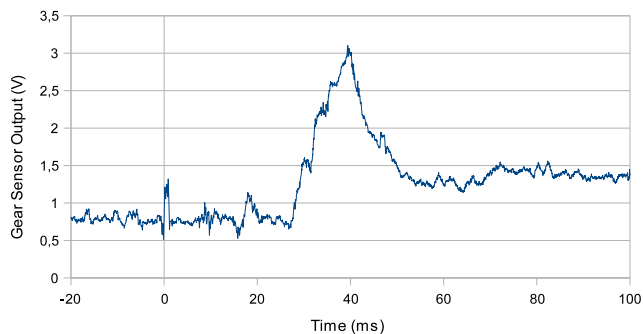


Fig. 12. Gear sensor output as a function of time: 2nd to 3rd shift

VI. CONCLUSIONS

The development of an innovative semiautomatic sequential gear for a Formula Student race car has been described in this paper. The main innovation is the adoption of linear voice coils as actuators of gear and clutch levers.

Since the electronic controlled gearbox is a multi-discipline device, a Matlab Simulink model to allow the co-simulation of the gear mechanics and the electric actuators was developed. The model developed here takes into account both the dynamic and the thermal characteristics of the actuators. The dynamic model is tailored to linear voice coil actuators but it can easily be adapted to rotary actuators. Model validation demonstrates that it is accurate and able to predict the actuator behavior with good precision. In addition, the thermal behavior of the voice coil actuator has been modeled, to take into account the heating caused by the Joule effect.

Then, the Gear Control Unit has been described in the main hardware and software components. The experimental results reported in Fig. 11 and Fig. 12 show that second-to-third up-shift lasts only 40 ms, a very good results if compared to other electric or hydroelectric solutions [18]-[20], or fast pneumatic systems with typical shift time of about 50 ms.

On the contrary, a weak point of the proposed system can be found in its weight, as the two voice coil actuators are heavier than those typically used in hydraulic or pneumatic systems. However, it should be noticed that the power-train section of the ET1 (including the engine, the gearbox and the GCU), weights about 38 kg, and the power-to-weight ratio obtained is comparable to that achieved by competitor cars that usually adopt 4-stroke 4-cylinder engines.

Finally, the functionality and reliability of the system were confirmed by the results of the last ATA 2008 race event, held in the Ferrari Circuit in Fiorano (Italy), to which the University of Pisa participated for the first time. The E-Team obtained remarkable results for a rookie team in this event, because it was 6th over 11 Italian teams and was one of the 14 teams (over 31 on the starting grid) that were able to complete the endurance race.

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